

Operation	Hardware type	Hardware length	Hops
Transactions ID			
Seconds		Flags	
Client IP address			
Your IP address			
Server IP address			
Router / Gateway IP address			
Client hardware address (16 bytes)			
Server host name (64 bytes)			
File (128 bytes)			
Operation			

Fig. 3: A DHCP message sent using UDP

Whereas, IPv6 supports a stateless auto configuration address. Thus, with IPv6, while moving from a subnet to another, a host can generate its own IP address. This is done by the host by adding its Media Access Control (MAC)/physical address to the subnet prefix.

IPv6 also supports multiple addresses for each host. Three types of address may be made: valid, deprecated or invalid.

- With valid address, new and existing communication may be done.
- With deprecated address, existing communication may be done.
- With invalid address, no communication is done.

When a host attempts to get an IP address, a DHCP discover message (Fig. 3) is broadcast over the physical network with broadcast IP address 255.255.255.25. The discover message is received by all routers and other hosts, which they ignore. Only the DHCP server or router replies. Other routers discard the discover message. This prevents the entire Internet from being flooded with discover message.

DHCP server in the network replies with DHCP offer message. The offer message provides an IP address and other information related to configuration. Every network need not have a DHCP server. A network may have DHCP relay agent, which requests the remote server for reply to be subsequently sent to the requesting host. If the requesting host receives a number of offers from a number of DHCP servers, it will accept one offer and acknowledge the same to the corresponding server, which then acknowledges the reply of the host. Then, the host uses the IP address for communication.

Mobile IP

Mobile IP is the real next extension of mobile voice communication. Advances in wireless network technology have immensely changed various parameters like process, method, perception, qual-

ity and characteristics of voice communication. Consequently, developments in wireless data protocols have paved the way for mobile IP for data communication.

What cellular technology, Global System for Mobile (GSM) communication or cellphone do for telephony, mobile IP will do for TCP/IP-based mode of data transport. IP Internet has become ubiquitous in wireline environments. Support of mobility is already available today in some forms using satellite access to an Internet service provider (ISP). But satellites provide access to just one ISP at a time, and only as long as the user is within the satellite's footprint.

Mobile IP provides a platform for carriers to offer assured and continuous access to the Internet or a private IP-based network. IP datagram is a packet with data plus header fields of different layers of the Internet architecture. This is like a letter that contains a message plus other data like sender and receiver names, addresses, pin codes, etc.

An IP datagram carries both source and destination addresses of data. Thus, if source or destination changes,

the connection becomes void until and unless the changed IP address is notified to all users.

In conventional wired Internet connectivity, when a user gets connected to the Internet, the point of attachment is made through an address known as IP address. This IP address is attached to a fixed plugged-in position.

In wireless connectivity, when a user is connected, he or she is connected to an IP address that is valid over the coverage area cell of wireless connectivity. The mobile user changes location dynamically with time. Thus, plugged-in fixed IP address of the wired Internet will be of no use to support connectivity to the mobile user.

On the other hand, there is no guarantee that mobility will be limited to the coverage area in case of wireless connectivity. IP address has to change when the mobile user moves away from own coverage or cell area to remain in connectivity.

One simple solution is to not assign a fixed IP address. There is a requirement to dynamically change the IP address. A new protocol called DHCP is made to resolve the stated issue. DHCP allows the intended user to obtain a temporary IP address dynamically. Like when one goes on deputation, one is allotted a temporary office/residential address.

This is a simple management solution. But a temporary address has two major disadvantages: temporary address notification and system upgradation. Other nodes cannot easily originate transmissions to the mobile node (MN) as these do not know about its new IP address. And each time a mobile

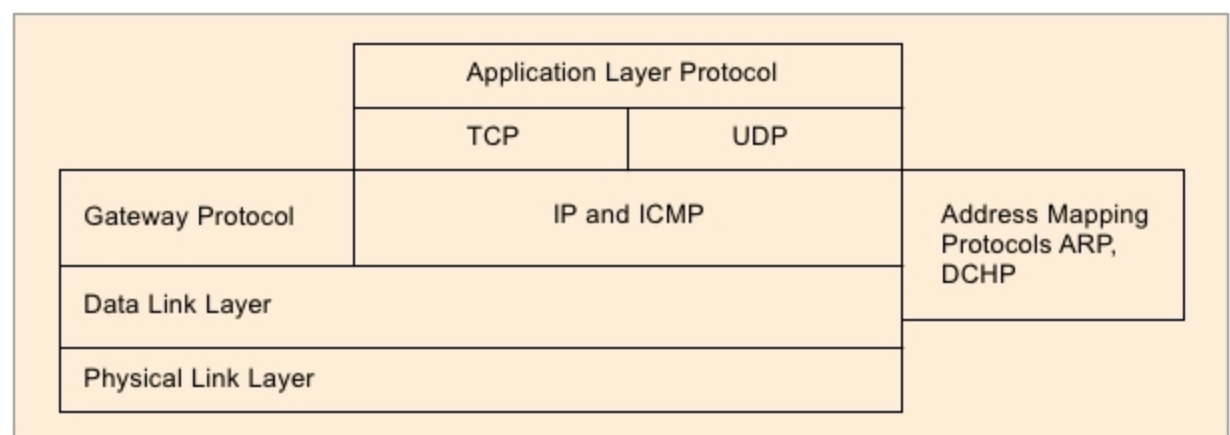


Fig. 4: TCP/IP (Internet) model [TCP = Transmission Control Protocol, UDP = Users' Datagram Protocol, ICMP = Internet Control Message Protocol, ARP = Address Resolution Protocol, DHCP = Dynamic Control Host Protocol]